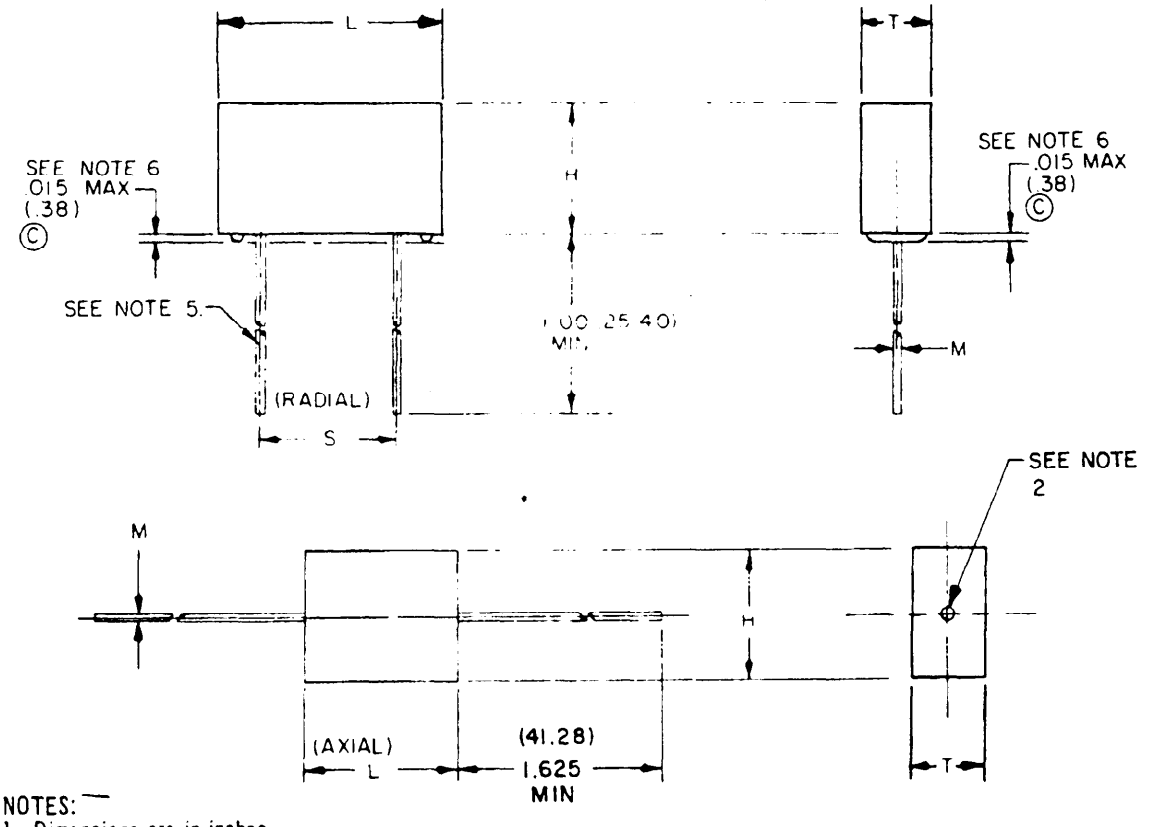
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC DIELECTRIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY, STYLE CFR04

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-55514.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



NOTES:
1. Dimensions are in inches.
2. Leads shall be of solid wire and located on centerline within ±.062 (1.57) but not less than .03 (.76) from edge of capacitor.
3. Millimeters are in parentheses.
4. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
5. Symmetry of centerline of leads with respect to end of case shall be within ±.010 (.25).
6. Stand-off not applicable to characteristics "M and Q"; optional for characteristic "S". ©

FIGURE 1. Style CFR04 capacitors.

© denotes changes.

FSC 5910

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TABLE I. Style CFR04 capacitor characteristics.

Type designation 1/ ©	Capacitance	Case size letter (see table II) for characteristics					
		M		Q			S
		Capacitance tolerance available G, J, K		Capacitance tolerance available F, G, J, K			Capacitance tolerance available F, G, J, K
		Rated voltage at 85° C					
	μF	200v	400v	100v	200v	400v	50v
CFR04---102--	.0010	-	D	-	-	D	A
CFR04---122--	.0012	-	D	-	-	D	A
CFR04---152--	.0015	-	D	-	-	D	A
CFR04---182--	.0018	-	D	-	-	D	A
CFR04---222--	.0022	-	D	-	-	D	A
CFR04---272--	.0027	-	D	-	-	D	A
CFR04---332--	.0033	-	D	-	-	D	A
CFR04---392--	.0039	-	D	-	-	D	A
CFR04---472--	.0047	-	D	-	-	D	A
CFR04---562--	.0056	-	D	-	D	E	A
CFR04---682--	.0068	-	D	-	D	E	A
CFR04---822--	.0082	-	D	-	D	E	A
CFR04---103--	.010	D	E	-	D	F	A
CFR04---123--	.012	D	E	-	D	F	B
CFR04---153--	.015	D	E	D	E	F	B
CFR04---183--	.018	D	F	D	E	F	B
CFR04---223--	.022	D	F	D	E	G	B
CFR04---273--	.027	D	F	E	F	G	B
CFR04---333--	.033	E	G	E	F	H	B
CFR04---393--	.039	E	G	E	G	H	B
CFR04---473--	.047	E	G	E	G	H	C
CFR04---563--	.056	F	H	F	G	H	C
CFR04---683--	.068	F	H	F	G	I	C
CFR04---823--	.082	F	H	F	H	I	C
CFR04---104--	.100	F	H	G	H	I	C
CFR04-SA124--	.12						J
CFR04-SA154--	.15			NOT			J
CFR04-SA184--	.18						K
CFR04-SA224--	.22			AVAILABLE IN			K
CFR04-SA274--	.27						L
CFR04-SA334--	.33			CHARACTERISTICS			L
CFR04-SA394--	.39						L ©
CFR04-SA474--	.47			M AND Q			L
CFR04-SA564--	.56						M
CFR04-SA684--	.68						M
CFR04-SA824--	.82						N
CFR04-SA105--	1.00						N

1/ The complete type designation will include additional symbols to indicate terminal type, characteristic and voltage rating (when applicable), capacitance tolerance, and failure rate level (M, P, R, or S).

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TABLE II. Style CFR04 dimensions

Case size	Dimension				
	H ±.030 (.76)	1/ L ±.015 (.38)	T ±.015 (.38)	S ±.005 (.13)	M ±.002 (.05)
A	.280 (7.11)	.500(12.70)	.125 (3.18)	.400(10.16)	.020(.51)
B	.375 (9.53)	.500(12.70)	.195 (4.95)	.400(10.16)	.025(.64)
C	.375 (9.53)	.600(15.24)	.195 (4.95)	.500(12.70)	.025(.64)
D	.471(11.96)	.500(12.70)	.234 (5.94)	.400(10.16)	.032(.81)
E	.411(10.44)	.800(20.32)	.178 (4.52)	.700(17.78)	.032(.81)
F	.475(12.07)	.800(20.32)	.244 (6.20)	.700(17.78)	.032(.81)
G	.531(13.49)	.800(20.32)	.291 (7.39)	.700(17.78)	.032(.81)
H	.569(14.45)	1.130(28.70)	.328 (8.33)	1.000(25.40)	.032(.81)
I	.971(24.66)	1.130(28.70)	.707(17.96)	1.000(25.40)	.032(.81)
J	.375 (9.53)	.600(15.24)	.295(15.24)	.500(12.70)	.025(.64)
K	.375 (9.53)	.750(17.05)	.500(12.70)	.650(16.51)	.025(.64)
L	.375 (9.53)	1.000(25.40)	.600(15.24)	.900(22.86)	.032(.81)
M	.375 (9.53)	1.500(38.10)	.600(15.24)	1.400(35.56)	.032(.81)
N	.550(13.97)	1.500(38.10)	.600(15.24)	1.400(35.56)	.032(.81)

1/ Add + 0.030 (.76) inch for overall length of axial-wire lead capacitors. ©

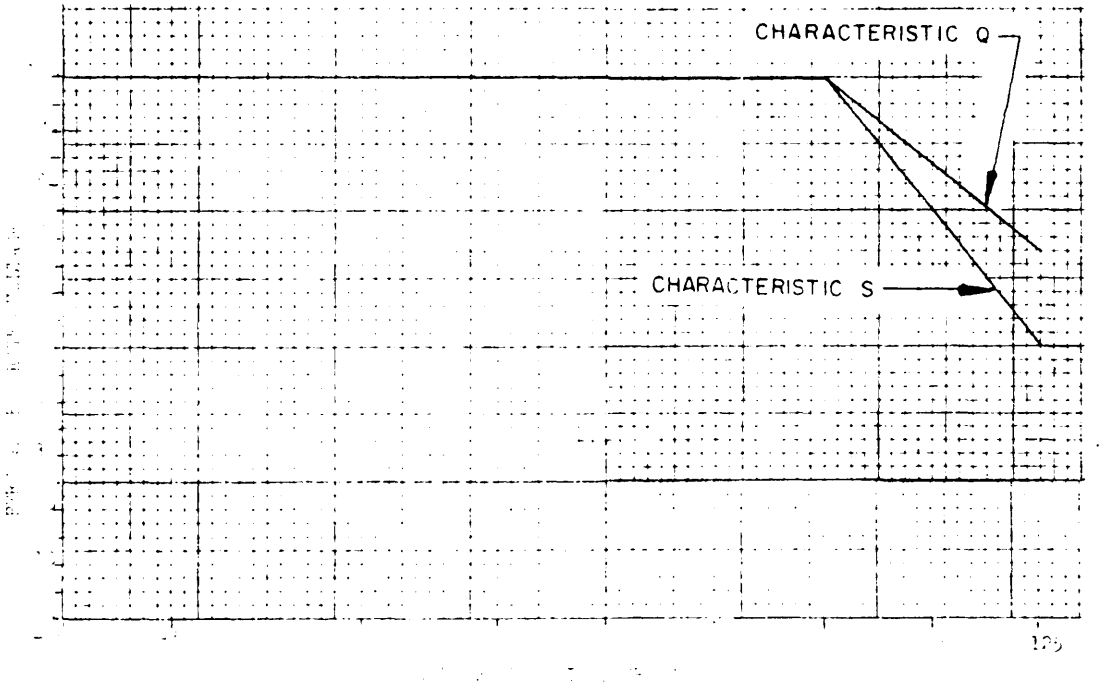


FIGURE 2. Voltage derating with temperature.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and tables I and II.
Terminals - Axial wire-lead (A) and radial wire-lead (R).
Characteristic - M, Q, or S as shown in table I.
Rated voltage - 50v (A), 100v (B), 200v (C), or 400v (E) as shown in table I. When properly voltage derated (see figure 2), will operate to 125°C.
Capacitance (nom) - See table I.
Capacitance tolerance - See table I.

Vibration, high frequency (direction and duration of motion, R-type terminals). - Three mutually perpendicular directions, perpendicular to the width, length, and thickness; 3 hours in each direction.

Failure rate level - M, P, R, or S.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army -
Navy - OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5910-1020-3)

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SPECIFICATION ANALYSIS SHEET

Form Approved
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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-55514/3C, CAPACITORS, FIXED, PLASTIC DIELECTRIC, DC, IN NONMETAL CASES, ESTABLISHED RELIABILITY, STYLE CFR04

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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Fort Monmouth, New Jersey 07703

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